

WORKING EXPERIENCE

- Abnormal Security** San Fransisco, CA / Remote from Ashburn, VA
Threat Intelligence Engineer *Sep 2021 – present*
 - Threat Intelligence Portal:** Build and deploy a threat intelligence portal from scratch with React.js, Django and PostgreSQL along with a set of libraries / APIs including Chart.js, Material UI, Pandas, Google APIs, Twilio API, RapidAPI, GPT API, etc to enable threat researchers with real-time scammer engagement, data query, filter and visualization while providing semi-auto message response and mule account collection feature
 - * **Scale:** Launch 90,000+ email security investigations and collect 40,000+ mule accounts within 1 year
 - * **Automation:** Automate 80% of threat actor communication and 75% of mule account collection
 - * **Traffic:** Deliver cutting edge research reports and blog posts to draw 70%+ traffic of the company website
 - * **Efficacy:** Help R&D team improve detection efficacy on various attack categories
 - * **Impact:** Send hundreds of real-time mule account notification to external partners including 40+ top financial institutes weekly to prevent potential financial losses of \$100,000+ and help drive POV to potential customers
 - Intelligence Augmentation:** Build an ETL pipeline with Python Scrapy crawlers / data processing scripts to collect auxiliary data from both OSINT and member-only resources to augment threat intelligence collection
 - * **Comprehension:** Add various data (e.g. domain age, IPs, validated bank codes, etc) to make intelligence collection more comprehensive to enable data analysis in different angles
 - * **Budget:** Save \$1,000+/yr by crawling public data, reducing API calls, building company-owned services
 - Internal Intel Sharing:** Build a S3-Databricks pipeline to sync collected threat intelligence and share raw data to internal teams along with dashboards and visualizations using SparkSQL to provide high level threat insight
 - * **Timeliness:** Perform hourly update enabling R&D to fine-tune filter rules to prevent latest attacks
 - * **Visualization:** Deliver 10+ dashboard / charts to show stakeholders most recent trend of threat landscape
 - * **Certification:** Databricks Certified Data Analyst Associate
- Agari Data** Foster City, CA / Remote from Ashburn, VA
DevOps Engineer / Research Intern *Feb 2020 – Sep 2021 / Jun 2018 – Dec 2019*
 - Advanced Intelligence Collection:** Build a honeypot with Flask, Bootstrap and S3 to trick scammers with fake wire confirmation and collect in-depth scammer intelligence; Build a Ruby on Rails App to manage each campaign and create an Elasticsearch cluster for efficient large-scale intelligence search
 - * **Scale:** Collect 300+ scammer profiles with 1TB+ intelligence in 2 years
 - * **Impact:** Uncover how Scattered Canary threat group uses stolen credentials to abuse COVID Pandemic Unemployment Assistant program and get stimulus check, results in 20+ media reports, raises company position in the email security market and prevents potential financial losses of hundreds of millions
 - Customer Intelligence Portal:** Build a customer portal with React.js, Node.js, Chart.js, REST API, Docker, Kubernetes to provide customers with real-time scammer data analysis, threat actor visualization and API query
 - * **Productized:** Create a one of a kind solution for financial institutes suffering from fraudulent account and financial loss by sharing with them timely intelligence and evidence for fraud investigation
 - * **Revenue:** Generate an ARR of \$200,000+ as the 4th product of the company
- Embry-Riddle Aeronautical University** Daytona Beach, FL
Teaching / Research Assistant *Aug 2017 – Dec 2019*
 - Java & Algorithm:** Teach Java programming, data structures and algorithms, provide tutoring hours & grading
 - Cybersecurity:** Conduct research including cloud & IoT security, privacy, phishing etc (see publications here)

EDUCATION

- Embry-Riddle Aeronautical University** Daytona Beach, FL
Ph.D. in Computer Science; *Jan 2016 – Dec 2019*
- Johns Hopkins University** Baltimore, MD
M.S. in Security Informatics; *Aug 2014 – Dec 2015*

PROGRAMMING SKILLS

- Languages:** Python, Ruby, Java, C\C++, React.js, Node.js, PHP, SQL, CSS, Matlab, L^AT_EX
- Libraries & Tools:** Tensorflow, Keras, SciPy, NumPy, Scrapy, Selenium, OpenCV, OpenAI, Django, Rails, Flask, Kubernetes, Drone, Vault, helm, AWS, GCP, Elasticsearch, Kibana, Docker, Nginx, Apache, Git, Metasploit, Nessus, Wiz, Synk, Air-crack, Wireshark